TECHNICAL REPORT

Source-Overlap-Controlled Transfer Learning for Still-Image Posture Recognition

A locked empirical study of scale, adaptation, classifier choice, ensemble diversity, external shift, attribution faithfulness, and bounded fault response

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Date	23 August 2026
Version	1.0.0
Status	Independent technical report

Repository
https://github.com/abdullahuseyinli-dot/human-activity-classification

Contents

Abstract

Still-image posture recognition is deceptively difficult to evaluate. Images that look different can be alternate crops, color variants, or neighboring frames from the same source; a high in-domain score can coexist with weak transfer; and visually plausible attribution maps can remain unchanged when their target class or model parameters are altered. This study addresses those problems in a four-class subset of POLAR containing sitting, standing, walking, and running. Before supervised fitting, a content audit quarantined 125 images belonging to 61 confirmed cross-split source-related components. All model, classifier, and ensemble decisions were then made on 13,285 development images. Nine neural fits and three frozen-feature probes were hash-verified before a one-time evaluation on 3,329 held-out test images.

The locked five-component probability ensemble reached 0.9399 macro-F1 (95% stratified bootstrap confidence interval [0.9312, 0.9481]) and 0.9456 accuracy. Its paired macro-F1 interval was positive against each component. A calibrated RBF support vector machine was the strongest individual component at 0.9274 macro-F1, while multinomial logistic regression was substantially smaller, faster to fit, and better calibrated. A fixed frozen-representation scale study increased from 0.8487 to 0.9150 validation macro-F1 between 242 and 9,958 training images. The largest class-level gains occurred for walking and standing, yet 95.6% of final ensemble errors still lay between adjacent posture or motion states.

The locked V-COCO transfer evaluation, preceded by a separate cross-dataset overlap audit, produced 0.6669 image-level macro-F1 for the collapsed ensemble and 0.6732 for adapted DINOv2-B. Post-hoc, non-selective analyses show that the gap combines several effects: smaller people, non-exclusive source actions forced into a three-class mapping, systematic standing-to-locomotion errors, and the inability of a full-frame model to vary predictions between people in the same image. Model disagreement was useful for detecting in-domain errors but much less informative under external shift, where many errors were unanimous. In the attribution audit, ConvNeXt Grad-CAM showed substantially greater target and parameter sensitivity than DINOv2-B integrated gradients. Bounded random bit flips caused few label changes, concentrated near ambiguous decision boundaries, but do not establish hardware or adversarial robustness.

The central result is methodological rather than architectural: strong in-domain performance became interpretable only when selection locking, source-overlap control, paired uncertainty, external evaluation, explanation sanity checks, and explicit limits were considered together. The study does not claim state of the art or deployment readiness.

1. Scope and contributions

The practical objective is to recognize four coarse human states from a single RGB image. The scientific objective is broader: to identify which apparent improvements remain credible after controlling selection, source overlap, random-seed variation, and evaluation uncertainty, and to determine where the resulting system fails.

The study asks six questions:

1. How does performance change across fixed training-set sizes?
2. When do full or partial backbone adaptation and frozen representations offer the

better accuracy-cost tradeoff?

3. Does a nonlinear SVM add useful separation after DINOv2 feature extraction?
4. Does a development-locked probability blend improve on its components?
5. Which error structures persist as data and model capacity increase?
6. Do in-domain accuracy, external transfer, attribution faithfulness, and bounded

fault stability support the same conclusions?

The work makes five evidence-backed contributions.

- It provides a source-overlap-controlled four-class POLAR benchmark with hash-locked manifests, a pre-fit quarantine, a one-time test gate, and paired uncertainty.

- It compares three adapted neural configurations and two frozen-feature classifiers within

one selection protocol, including an explicit efficiency analysis.

- It reports evidence about additional data separately from regularization, adaptation,

seed averaging, and probability blending.

- It evaluates the locked system under a second dataset, explanation sanity checks,

and exact bounded bit-flip interventions rather than relying on in-domain accuracy alone.

- It assembles portable metrics, exclusion records, selection locks, provenance,

hashes, and figure-generation code while keeping rights-sensitive and bulky data local.

The contribution is a controlled empirical account, not a new network architecture. No claim is made that the four-class score is directly comparable with published nine-class POLAR results, subject-independent protocols, V-COCO role average precision, video recognition, or sensor-based activity recognition.

2. Related work

2.1 Still-image posture and action recognition

POLAR was introduced as a posture-level action recognition dataset spanning nine classes and substantial variation in person scale, scene, and viewpoint [1, 2]. The present study uses only four labels and applies a new content-based split audit. Consequently, the original paper and this report address related data but do not define identical benchmarks.

V-COCO extends COCO with person-centric action annotations and was designed for visual semantic role labeling rather than mutually exclusive posture classification [3]. Using V-COCO here therefore creates a deliberately imperfect external stress test. Source actions are mapped to sitting, standing, or a collapsed walking/running class, and the report preserves the semantic mismatch rather than presenting the result as a standard V-COCO benchmark.

Recent pose-based industrial action-recognition work by Huang et al. combines an improved YOLO-Pose detector with recurrent modeling and evaluates it on the privately constructed 12-class SKPose dataset [4]. The study is useful methodological context, but differences in labels, temporal construction, splits, dataset access, and metrics prevent a direct ranking. This report confines quantitative comparisons to candidates evaluated on the same locked rows.

2.2 Transfer representations and adaptation

DINOv2 learns general-purpose visual features without task-specific supervision and has shown strong transfer across image recognition tasks [5]. ConvNeXt modernizes a convolutional design using training and architectural choices associated with vision transformers [6]. Their different inductive biases motivate their joint use here: DINOv2 models operate on a person-centered crop with context, whereas ConvNeXt retains the full frame.

ImageNet performance often correlates with transfer performance, but transfer quality also depends on the target task, the representation layer, and the adaptation procedure [7]. Fine-tuning can improve in-domain fit while altering features in ways that do not necessarily help distribution shift [8]. Intermediate or multilayer features can also outperform a final-layer-only representation for downstream classification [9]. These findings motivate the controlled comparison among full adaptation, partial adaptation, a frozen multilayer representation, and simple downstream classifiers.

2.3 Classifiers, calibration, and ensembles

The support vector machine remains a strong nonlinear classifier when a useful representation is already available [10]. The relevant engineering question is not whether an RBF decision boundary can improve a score in isolation, but whether the increment justifies calibration cost, support-vector storage, and inference complexity. This study therefore retains both a calibrated RBF SVM and a multinomial logistic probe.

Modern neural classifiers can be poorly calibrated even when their accuracy is high [11]. Probability averaging across independently trained models often improves both predictive accuracy and uncertainty quality [12]. The present ensemble combines model families, input views, adaptation levels, and classifier heads, with weights selected only from development predictions.

2.4 Evaluation under shift and explanation audits

Near-duplicate images can materially inflate held-out image recognition results when related content crosses partitions [13]. Domain-generalization research has likewise shown how easily model-selection choices confound conclusions about robustness [14]. Accuracy under natural distribution shift often tracks, but does not perfectly follow, in-domain accuracy [15, 16]. These observations motivate a source audit, a locked test gate, and an external dataset rather than a single random split.

Integrated gradients and Grad-CAM are widely used to visualize model evidence [17, 18]. Visual plausibility alone is insufficient: saliency methods should respond to changes in the target function and learned parameters [19], and perturbation metrics require appropriate controls [20, 21]. This study reports target sensitivity, parameter randomization, deletion and insertion behavior, and person-region localization separately. It does not treat an attribution map as an explanation merely because it resembles a person mask.

Finally, targeted bit manipulation can severely damage quantized neural networks under adversarial fault models [22, 23]. The fault experiment in this repository is much narrower. It measures local stability to declared random bit flips at specific inputs and classifier matrices; it is neither a security evaluation nor a hardware reliability claim.

3. Data and source-overlap controls

3.1 Primary task

The source dataset contains 35,324 annotated images across nine posture-level actions. The locked task retains images labeled sit, stand, walk, or run and maps them to sitting, standing, walking, and running. Walking and running are also collapsed for a secondary three-class transfer task.

Clean split	Sitting	Standing	Walking	Running	Total
Train	3,021	2,720	1,994	2,223	9,958
Validation	1,000	967	648	712	3,327
Test	1,034	921	638	736	3,329
Total	5,055	4,608	3,280	3,671	16,614

The class distribution is not uniform, so macro-F1 is the primary metric. Accuracy, balanced accuracy, weighted F1, per-class precision and recall, log loss, multiclass Brier score, and 15-bin expected calibration error (ECE) provide complementary views.

3.2 Content audit and quarantine

The source split was not assumed to be independent merely because filenames differed. The audit first checked byte hashes and then retrieved visually related candidates. Candidate cross-split pairs had to satisfy both a 64-bit perceptual-hash distance of at most six and a resized grayscale correlation of at least 0.90. Frozen DINOv2 embeddings proposed additional candidates, but embedding proximity alone was never used as an exclusion rule.

The audit confirmed 64 source-related cross-split pairs forming 61 connected components. Every image belonging to a confirmed component was quarantined, removing 125 images before supervised fitting. The locked clean manifest contains 16,614 rows. Quarantined test content was not reassigned to development.

This procedure controls the specific relationships found by the declared retrieval and confirmation rules. It does not prove complete leakage removal. Subject and capture session identifiers were not available, and source relationships outside the retrieval thresholds may remain. For this reason, the report uses the terms source-overlap-controlled and leakage-audited rather than leakage-free.

3.3 External data

V-COCO train and validation annotations provide a separate three-class audit. The local manifest contains 6,640 mapped person annotations in 4,123 images. Image-level evaluation uses 3,761 images whose relevant people agree on one mapped label. Images with mixed mapped person labels remain in the person-level analysis.

The fixed mapping assigns `sit` to sitting, `stand` to standing, and either `walk` or `run` to the combined walking/running label. Locomotion takes precedence when `stand` co-occurs with `walk` or `run`; rows containing only `{sit, stand}` are excluded. Image-level probabilities are averaged across mapped people only when those people share one mapped label; all 6,640 mapped rows remain in the person-level analysis.

A separate cross-dataset audit compared all 16,614 clean POLAR images against all 4,123 V-COCO images. It found no byte-identical matches, no perceptual candidates within the declared threshold, and no confirmed source-related pairs. This excludes the detected forms of direct content overlap; it does not make the domains semantically equivalent.

4. Locked experimental design

4.1 Selection boundary

The official training split fitted candidates, and the validation split made every modeling decision. Search proceeded in bounded stages:

- frozen probes screened label mappings, full-frame and person-context views, and

class-balance choices;

- adaptation runs compared head-only, last-stage or top-four-block, and full-backbone

training;

- controlled regularization runs examined dropout, augmentation, MixUp [27], label

smoothing, class weighting, weight decay, and random erasing;

- seeds 42, 52, and 62 confirmed competitive settings;
- DINOv2-B entered through a predeclared capacity extension;
- classifier settings and ensemble weights were selected from development

predictions only.

Macro-F1 was the primary selector. Differences smaller than 0.002 were treated as a practical tie. Declared tie breakers considered seed variability, log loss, ECE, and cost. Training failures were retained in the run ledger rather than removed.

After selection, the clean training and validation partitions were combined for the final fits. Nine neural checkpoints and three classifier probes were checked against their request, source, and artifact hashes. The test cache then opened once. No test score changed a model, weight, threshold, epoch count, or attribution method.

4.2 Neural components

Model	View	Adaptation	Augmentation	Dropout	Final epochs
ConvNeXt-S	Full frame	Full backbone, layer decay 0.70	Mild	0.10	12
DINOv2-S	Person plus 25% context	Full backbone, layer decay 0.75	Moderate	0.10	12
DINOv2-B	Person plus 25% context	Top four blocks	Mild	0.10	7

Each neural component averages probabilities across three independently seeded final fits. The fixed epoch counts are the median best epochs from the relevant confirmation runs. The final fit ledger records a combined runtime of 11,557 seconds for the nine neural fits and three probes.

4.3 Frozen DINOv2-B representation and probes

The frozen representation concatenates normalized class tokens from the last four transformer layers and the normalized mean patch token from the final layer. Features from a full-frame view and a person crop with 10% context are concatenated, producing 7,680 dimensions.

Two downstream classifiers were retained:

- standardized multinomial logistic regression with C = 0.001 and balanced class

weights;

- an RBF SVM with C = 10 and gamma = 1/7,680, followed by five-fold sigmoid

calibration.

The RBF hyperparameters were transferred from the declared development screen rather than retuned after constructing the final representation.

4.4 Locked probability blend

Component	Weight
ConvNeXt-S, full adaptation	0.20
DINOv2-S, full adaptation with moderate augmentation	0.15
DINOv2-B, top-four-block adaptation	0.25
DINOv2-B multilayer logistic probe	0.20
DINOv2-B multilayer calibrated RBF SVM	0.20

The weights are an arithmetic blend of class probabilities. They were selected on validation predictions in increments of 0.05 using macro-F1, followed by log loss and ECE. No subsequent post-hoc analysis alters this blend.

4.5 Uncertainty and analysis roles

Locked estimates use 10,000 class-stratified bootstrap resamples with seed 20260822. Candidate comparisons use paired resamples of the same rows. The external image-level audit uses the corresponding locked external bootstrap.

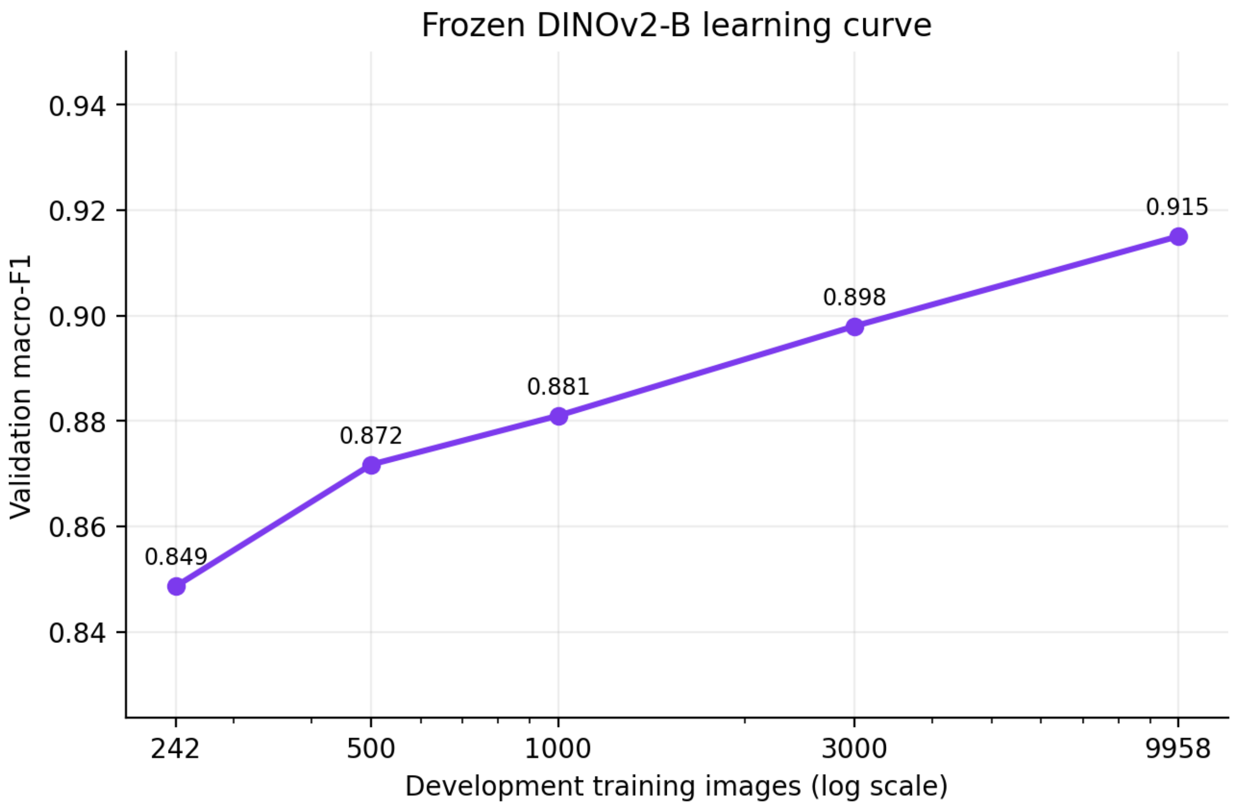
Analyses added after the test evaluation are labeled post hoc and non-selective. They reuse fixed predictions to explain observed behavior; they do not choose a replacement model, modify the primary result, or turn descriptive correlations into confirmatory claims. Post-hoc intervals reported below use 10,000 resamples with analysis-specific fixed seeds. Unadjusted exploratory p-values, where reported in portable files, are descriptive only.

5. Development evidence

5.1 Training-set scale

The fixed scale experiment uses frozen DINOv2-B full-frame and 10%-context features with standardized multinomial logistic regression (C = 0.01, no class weighting) across nested, class-stratified training subsets. Validation macro-F1 rose monotonically from 0.8487 with 242 rows to 0.9150 with all 9,958 training rows.

Training rows	Validation macro-F1
242	0.8487
500	0.8718
1,000	0.8810
3,000	0.8980
9,958	0.9150



Fixed training-scale curve

The exported run table contains three records per size, but those records share the same subset hash, output, and macro-F1 at each size. They are repeated executions of one deterministic nested subset curve, not three independent sample draws. The evidence therefore supports a five-point descriptive curve, not a variance estimate or a scaling law.

Post-hoc, non-selective decomposition shows that the gain from 242 to 9,958 rows was largest for walking (+0.0990 class F1) and standing (+0.0809), followed by running (+0.0478) and sitting (+0.0378). Walking and standing account for approximately 67.7% of the summed class-F1 improvement. More data reduced the number of errors, but it did not change their basic topology: adjacent-state errors represented 88.1% of errors at 242 rows and 90.7% at 9,958 rows. Within this fixed pipeline, the curve is consistent with improved boundary placement as labeled rows increase, while the underlying still-image ambiguity remains.

5.2 Adaptation and capacity

A post-hoc, non-selective synthesis of the development runs does not support a rule that more trainable parameters always produce a better model. For DINOv2-B at seed 42, full adaptation reached 0.9245 validation macro-F1, while top-four-block adaptation reached 0.9240. The top-four variant trained 28.4 million of 86.6 million parameters, approximately 32.8% of the model's parameters, and completed about 20.9% faster. This is a practical tie with a substantial resource difference.

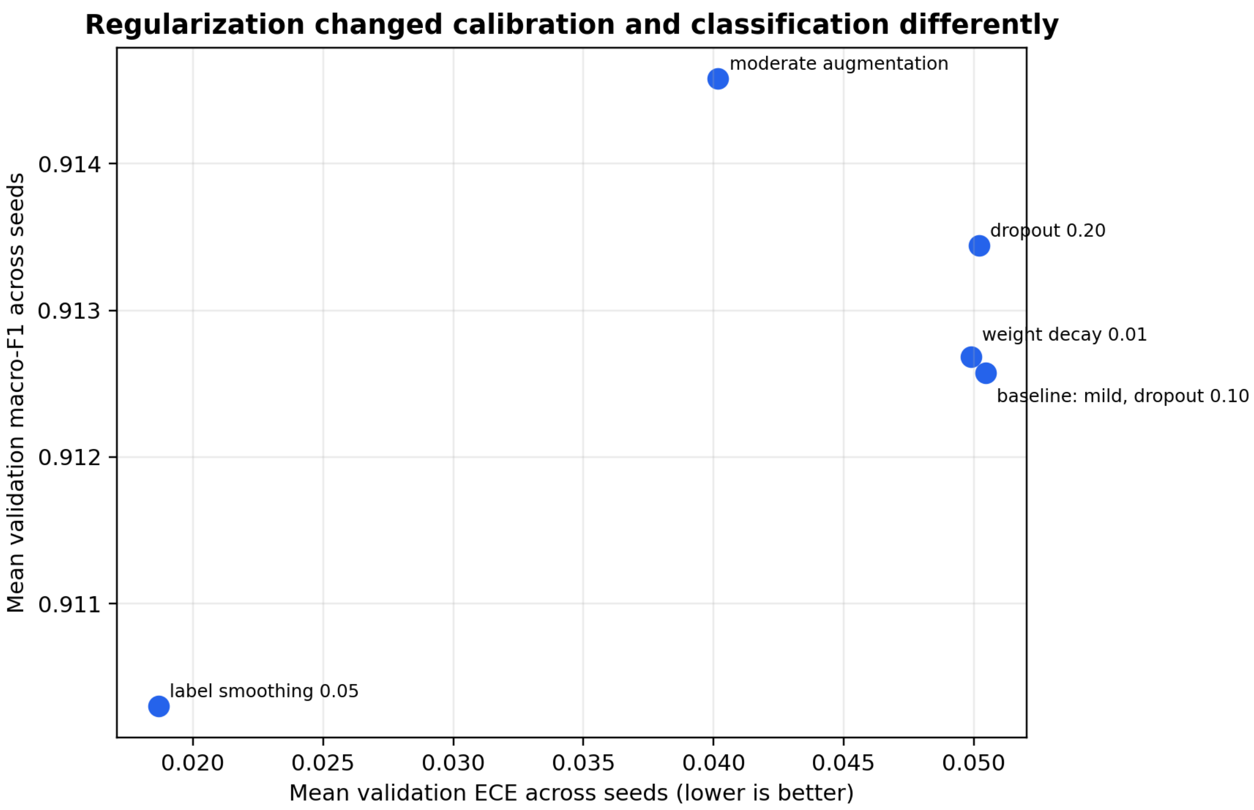
DINOv2-S behaved differently. Full adaptation reached 0.9184 at seed 42, compared with 0.9168 for top-four adaptation. The partial variant also ran longer in that comparison because its selected epoch was later. ConvNeXt-S showed a clearer dependence on adaptation depth: full adaptation reached 0.8920 validation macro-F1, compared with 0.8643 for last-stage training and 0.8059 for a head-only fit.

These are pipeline-specific observations. They do not isolate architecture from input view, schedule, or pretrained representation. They show that trainable fraction by itself is an inadequate predictor of accuracy or wall-clock cost.

5.3 Post-hoc, non-selective regularization synthesis

The underlying DINOv2-S regularization variants were evaluated in the development stage. The following post-hoc synthesis compares the five specifications with complete three-seed evidence.

Variant	Mean macro-F1	Seed SD	Mean log loss	Mean ECE
Mild augmentation, dropout 0.10	0.9126	0.0055	0.3227	0.0505
Moderate augmentation	**0.9146**	0.0049	0.2710	0.0402
Dropout 0.20	0.9134	0.0061	0.3282	0.0502
Label smoothing 0.05	0.9103	**0.0035**	**0.2579**	**0.0187**
Weight decay 0.01	0.9127	0.0055	0.3220	0.0499



Regularization tradeoffs

Moderate augmentation gave the highest mean macro-F1 and improved probability quality relative to the baseline. Label smoothing [25] sacrificed about 0.23 macro-F1 percentage points relative to the baseline while sharply improving ECE and log loss. It is therefore inaccurate to classify every lower-F1 intervention as a failed regularizer; the ranking depends on whether the objective is classification, calibration, stability, or cost.

There is an important DINOv2-S selection distinction. The individual multi-seed selector chose the top-four variant because it lay within the 0.002 practical-tie band and had the lowest seed standard deviation. The later development-only blend search, however, assigned that DINOv2-S top-four candidate zero weight and assigned 0.15 to the moderate full-adaptation candidate. The final ensemble therefore uses the moderate variant because it contributed to the system-level validation blend, not because it won the standalone confirmation ranking.

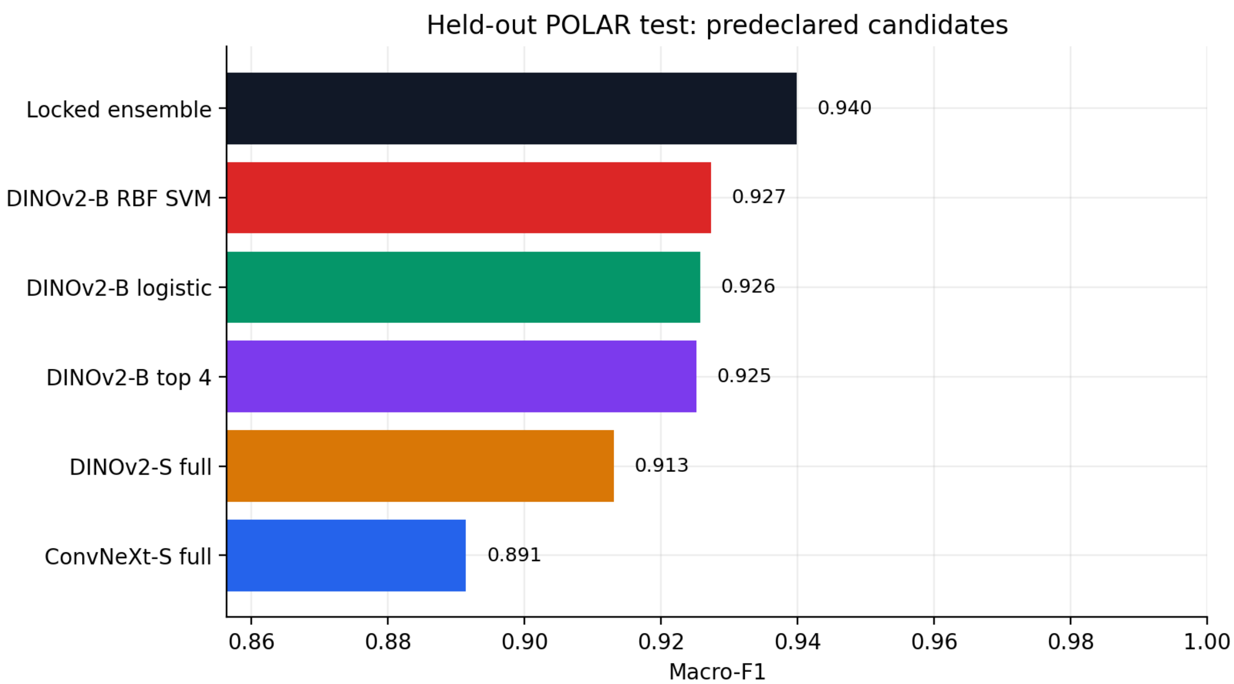
5.4 Post-hoc, non-selective seed-averaging synthesis

Averaging three final seed probabilities produced small, consistent gains over the mean of individual seed macro-F1 values: approximately +0.0013 for ConvNeXt-S, +0.0010 for DINOv2-B, and +0.0019 for DINOv2-S. Log loss and ECE also improved. These increments are modest compared with the scale effect, but they are low-risk once multiple final fits already exist.

6. Locked in-domain evaluation

6.1 Primary result

Candidate	Macro-F1	95% CI	Accuracy	Log loss	ECE
Locked ensemble	**0.9399**	[0.9312, 0.9481]	**0.9456**	**0.1564**	0.0291
DINOv2-B multilayer RBF SVM	0.9274	[0.9179, 0.9362]	0.9342	0.2280	0.0269
DINOv2-B multilayer logistic	0.9258	[0.9164, 0.9348]	0.9324	0.1764	**0.0133**
DINOv2-B top four	0.9252	[0.9156, 0.9343]	0.9327	0.2173	0.0420
DINOv2-S full	0.9131	[0.9028, 0.9226]	0.9210	0.2389	0.0298
ConvNeXt-S full	0.8914	[0.8803, 0.9020]	0.8994	0.3081	0.0429



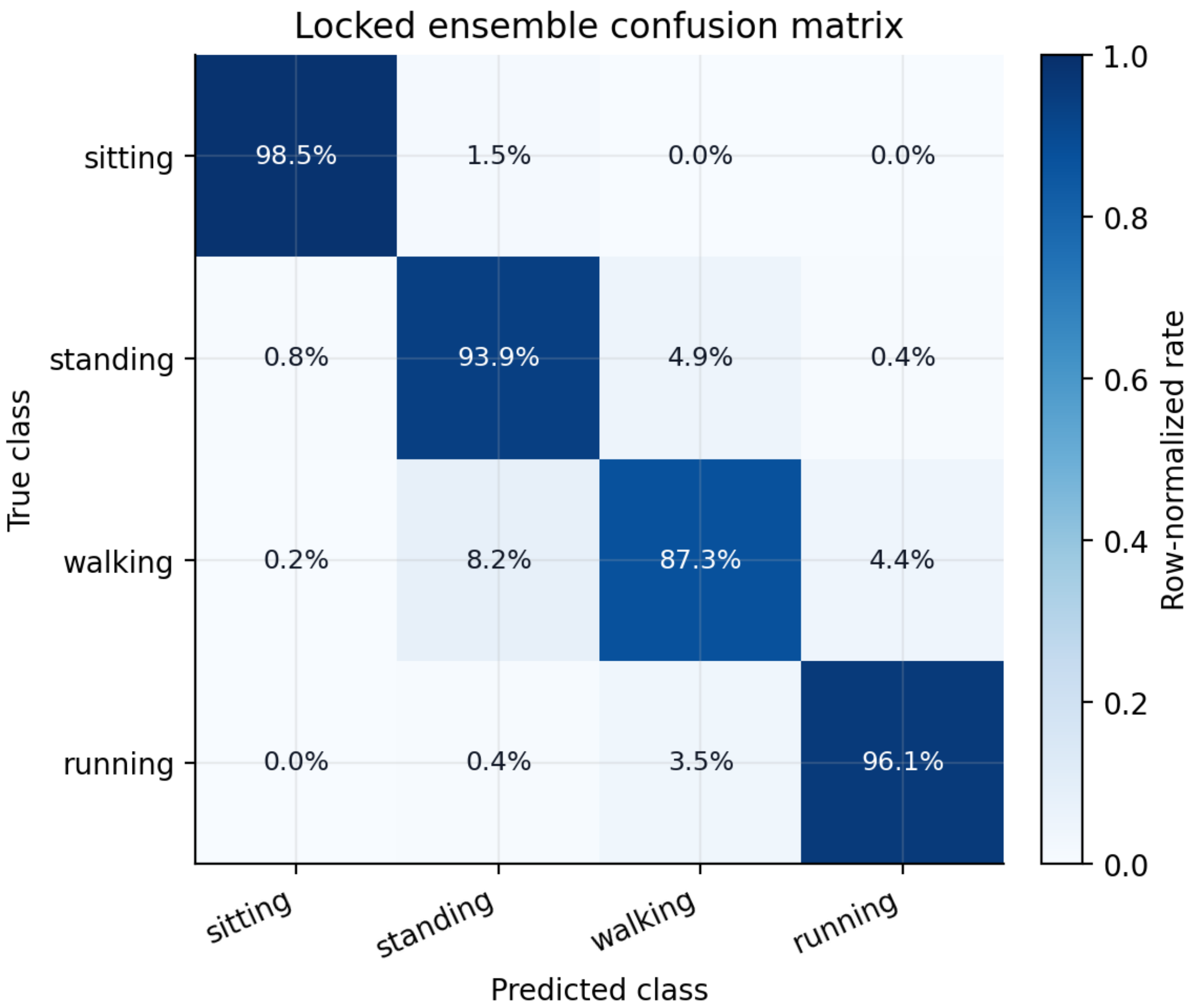
Locked held-out comparison

The ensemble's paired macro-F1 gain was +0.0484 over ConvNeXt-S, +0.0268 over DINOv2-S, +0.0147 over adapted DINOv2-B, +0.0141 over logistic regression, and +0.0125 over the RBF SVM. All five locked paired 95% intervals were above zero. The narrowest comparison was against the RBF SVM, with interval [0.0065, 0.0186].

For the secondary task, collapsing walking and running after inference produced 0.9611 macro-F1 and 0.9622 accuracy. A direct three-class logistic probe reached 0.9531 macro-F1. This secondary comparison was locked and uses the same test rows; it does not establish performance on unrelated three-class activity taxonomies.

6.2 Per-class behavior

Class	Precision	Recall	F1	Support
Sitting	0.992	0.985	0.989	1,034
Standing	0.925	0.939	0.932	921
Walking	0.887	0.873	0.880	638
Running	0.957	0.961	0.959	736



Locked ensemble confusion matrix

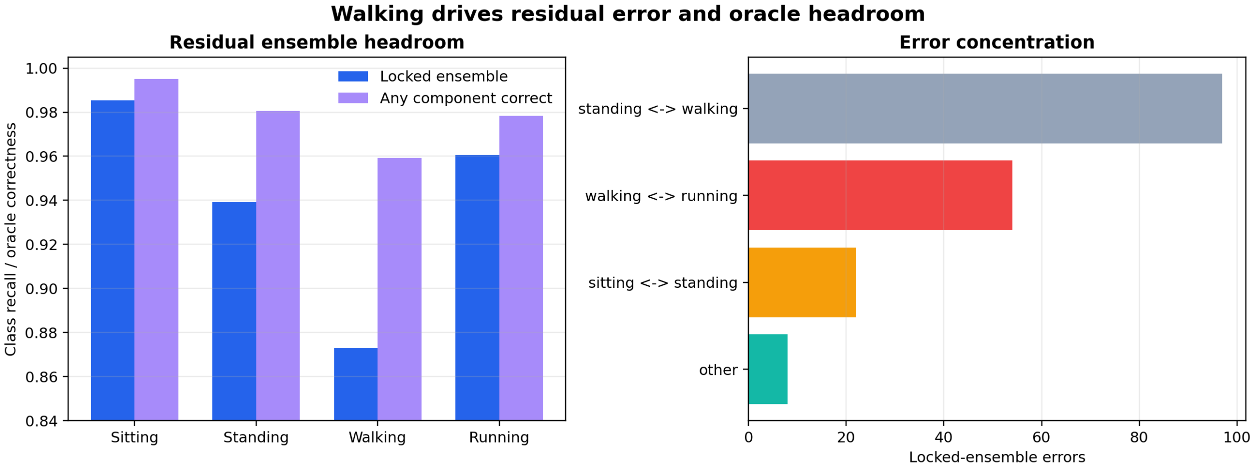
Walking is the limiting class. Of 638 walking examples, 52 were predicted as standing and 28 as running. A post-hoc, non-selective interpretation is that sitting and running form more visually distinctive endpoints in this descriptive label ordering.

6.3 Post-hoc, non-selective error topology

The four labels have a natural descriptive order:

sitting <-> standing <-> walking <-> running.

This order was not used as a loss function or selection constraint. It is applied after evaluation to summarize the confusion matrix. Of the ensemble's 181 errors, 173 (95.6%) occur between adjacent states. Standing and walking account for 97 errors (53.6% of all errors), walking and running for 54 (29.8%), and sitting and standing for 22 (12.2%). Only eight errors cross more than one boundary.



Post-hoc error structure

The result suggests that the remaining difficulty is structured rather than random. A still frame can show a person between gait phases, standing immediately before motion, or partially seated. The same adjacent-state concentration appears along the fixed scale curve and in the highest-rate fault diagnostic. This recurring pattern supports future work on ordinal objectives, temporal evidence, or uncertainty around transition states. It does not prove that the source labels themselves form a strict ordinal variable.

6.4 Post-hoc, non-selective ensemble structure

At least one of the five components was correct on 98.05% of test images, compared with 94.56% accuracy for the realizable fixed blend. This 3.48 percentage-point oracle gap is an upper-bound diagnostic: it assumes knowledge of the true label for choosing a component and is not a deployable result.

Component disagreement occurred on 15.74% of test images. It was highest for walking at 33.07%, compared with 18.57% for standing, 11.41% for running, and 5.61% for sitting. The oracle and disagreement results show that the net improvement is an aggregate consequence of complementary errors, not a guarantee that blending improves every row.

The exact validation-selected weights are not uniquely important. An equal-weight average reached 0.9395 macro-F1, only 0.00034 below the locked 0.9399 result, and agreed with it on 99.76% of predictions. The post-hoc paired interval for locked minus equal weights is [-0.0012, 0.0020]. In contrast, removing any one component reduced macro-F1 by approximately 0.0019 to 0.0029. Taken together, these checks indicate that the component set is more consequential than fine adjustment of the five weights.

6.5 Post-hoc, non-selective selective prediction

Maximum probability is informative in-domain. The locked ensemble's confidence had an area under the ROC curve of 0.9225 for distinguishing correct from incorrect predictions. Retaining the 90% most confident test rows yielded 0.9770 selective accuracy, compared with 0.9456 at full coverage.

This analysis is descriptive and does not define an operational rejection threshold. Coverage was examined after the test result, no cost for abstention was specified, and calibration can shift across domains. The external analysis below confirms that this caution is necessary.

7. Classifier and system tradeoffs

7.1 Linear and nonlinear frozen-feature probes

Probe	Test macro-F1	Test log loss	External image macro-F1	Fit time	Serialized probe
Multinomial logistic	0.9258	**0.1764**	0.6392	**13.9 s**	**0.4 MB**
Calibrated RBF SVM	**0.9274**	0.2280	**0.6504**	60.4 min	870.9 MB

The RBF SVM adds 0.0016 macro-F1 in-domain. Its post-hoc paired interval against logistic regression is [-0.0051, 0.0085], so the evidence does not establish a statistically resolved advantage. The RBF probe also requires five calibrated fits, approximately 5,038 support vectors per fold, and over 2,000 times more serialized classifier storage.

The storage figures in the table cover only the downstream classifiers. Both require the same DINOv2-B feature extractor, approximately 346 MB of shared backbone weights, as well as the cost of two-view feature extraction. Omitting that common cost would make the logistic alternative look like a standalone sub-megabyte vision system, which it is not.

Logistic regression is the stronger default where calibration, fit time, storage, or maintainability dominates. The RBF SVM remains useful as a research component because its nonlinear boundary contributes a different probability surface to the blend.

7.2 Post-hoc, non-selective adapted-versus-frozen pipeline comparison

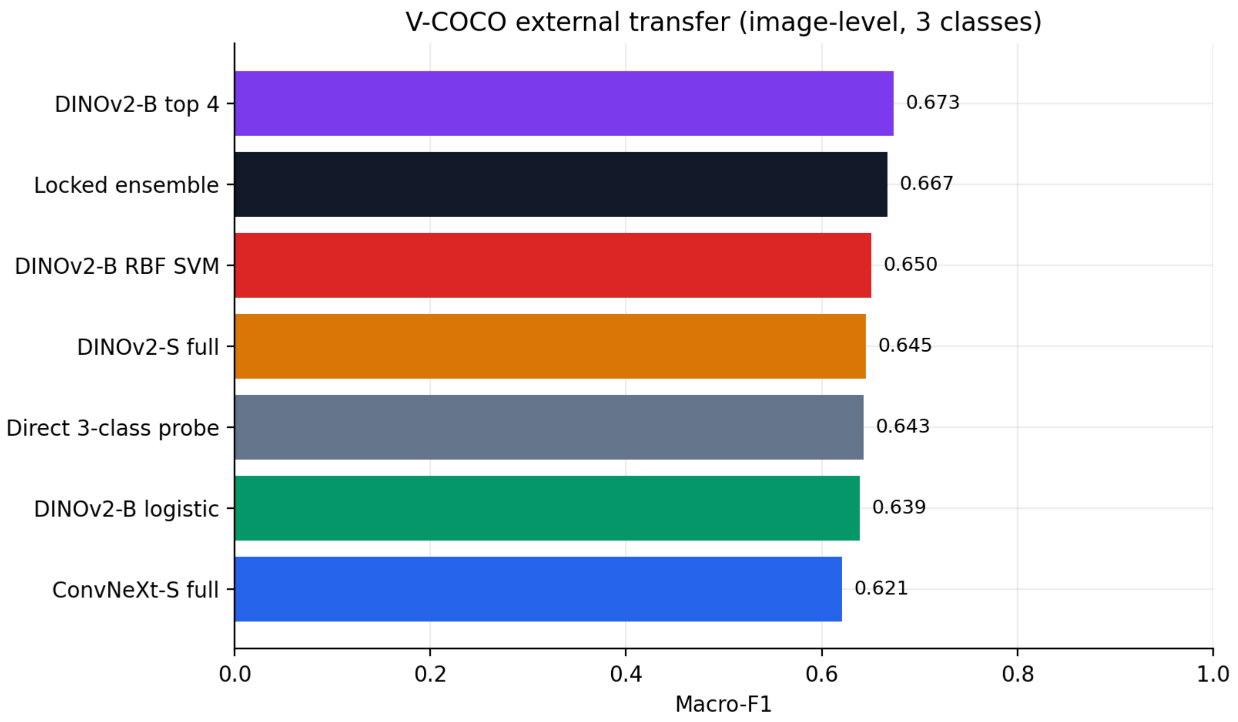
Adapted DINOv2-B, logistic regression, and the RBF SVM are effectively tied in-domain at the resolution of their paired intervals: [-0.0097, 0.0088] for adapted minus logistic and [-0.0114, 0.0070] for adapted minus RBF. On V-COCO images, adapted DINOv2-B exceeds the logistic probe by 0.0340 macro-F1 (95% interval [0.0242, 0.0442]) and the RBF probe by 0.0228 (95% interval [0.0125, 0.0333]).

This contrast is suggestive but not an adaptation-only causal comparison. The adapted model uses a person crop with 25% context and a learned classification head, whereas the frozen probes concatenate full-frame and 10%-context features from several layers. The result supports a narrower hypothesis: the adapted DINOv2-B pipeline preserves more useful behavior under this particular external shift, even though the three pipelines are nearly indistinguishable on POLAR.

8. External transfer

8.1 Locked V-COCO results

V-COCO image-level candidate	Macro-F1	95% CI	Accuracy	Log loss
DINOv2-B top four	**0.6732**	[0.6609, 0.6849]	**0.6751**	1.7509
Locked ensemble, collapsed	0.6669	[0.6550, 0.6787]	0.6663	**1.2512**
DINOv2-B multilayer RBF	0.6504	[0.6380, 0.6623]	0.6453	2.0801
DINOv2-S full	0.6455	[0.6329, 0.6579]	0.6472	1.6232
Direct three-class probe	0.6430	[0.6309, 0.6545]	0.6405	1.2775
DINOv2-B multilayer logistic	0.6392	[0.6274, 0.6507]	0.6363	1.4022
ConvNeXt-S full	0.6208	[0.6073, 0.6337]	0.6272	1.8414



External transfer results

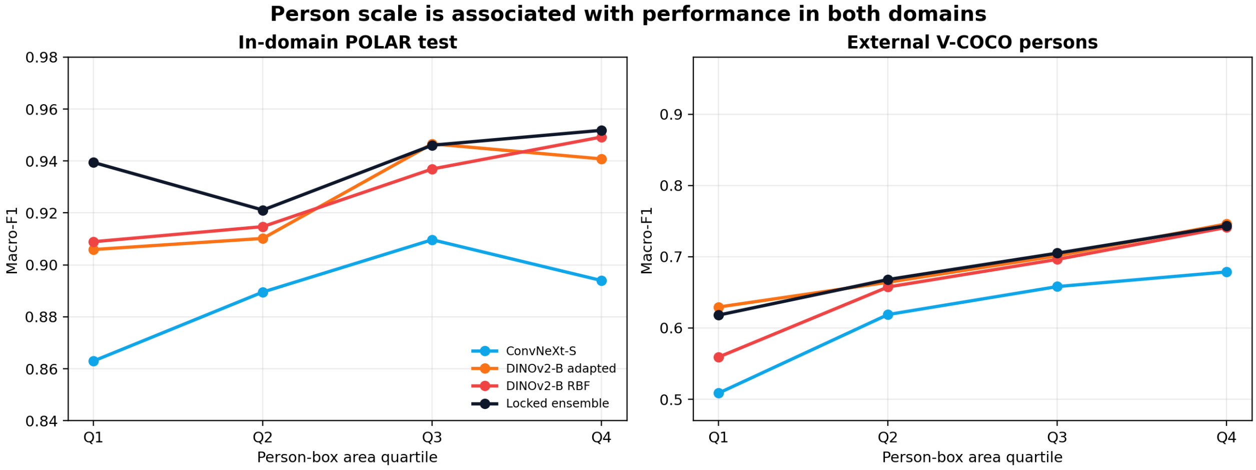
The collapsed in-domain ensemble reaches 0.9611 macro-F1, whereas the same locked probabilities yield 0.6669 on V-COCO images. The 0.2942 gap is the most consequential negative result in the study. It prevents the in-domain score from being interpreted as evidence of general deployment performance.

The external ranking also changes. Adapted DINOv2-B has the highest image-level macro-F1, exceeding the ensemble by 0.0063, but their post-hoc paired interval [-0.0016, 0.0141] includes zero. The ensemble has substantially better log loss and ECE than adapted DINOv2-B. There is therefore no single external winner across discrimination and probability quality.

At person level, adapted DINOv2-B reaches 0.6853 macro-F1 and the ensemble reaches 0.6840. Person-level evaluation retains all 6,640 mapped people, including those from images with mixed person labels. These values cannot be compared directly with standard V-COCO role average precision because the task, mapping, unit of analysis, and metric differ.

8.2 Post-hoc, non-selective person-scale analysis

Person boxes are materially smaller in V-COCO. Across the mapped data, the median person-box area is 0.145 of image area in V-COCO and 0.296 in POLAR, a ratio of 0.492. Half of V-COCO people are smaller than the first POLAR box-area quartile threshold. Class-specific median ratios are 0.611 for sitting, 0.405 for standing, and 0.488 for walking/running.



Post-hoc person-scale analysis

On POLAR, the locked ensemble macro-F1 is 0.9394 in the smallest-box quartile and 0.9518 in the largest. The RBF probe changes more sharply, from 0.9089 to 0.9492. The ensemble's 0.0305 advantage over the RBF probe in the smallest quartile has a positive post-hoc paired interval [0.0168, 0.0452]. The unadjusted descriptive Spearman correlation between box-area fraction and binary prediction correctness is 0.066 for the ensemble and 0.085 to 0.139 for individual components. This is consistent with, but does not prove, reduced size dependence through blending.

V-COCO person-level ensemble macro-F1 rises from 0.6181 in the smallest external quartile to 0.7435 in the largest. Even the largest-person group remains far below the in-domain result. Person scale is therefore one contributor to the transfer gap, not a complete explanation.

As a check for annotation geometry signal, a post-hoc fixed multinomial logistic baseline used only five ground-truth annotation features: log box area, log box aspect ratio, box-center x and y coordinates, and log image aspect ratio. It reached 0.4310 macro-F1 on POLAR test (95% interval [0.4140, 0.4478]), above the majority-class descriptive reference but far below the visual models. When transferred without retuning to V-COCO people, it reached 0.3073 macro-F1 and predicted the locomotion class for 71.3% of rows. Geometry carries label-correlated information within POLAR, but its relationship to the mapped classes shifts across datasets. Because the baseline requires ground-truth boxes, it is a diagnostic rather than a deployable method.

8.3 Post-hoc, non-selective semantic-mapping analysis

V-COCO actions are not mutually exclusive. Of 1,261 person rows mapped to walking/running, 1,174 (93.1%) also carry the source action stand. At image level, 584 of 621 locomotion images (94.0%) have at least one relevant person with a stand co-tag, and 562 (90.5%) have stand tags for every relevant person.

The forced three-class mapping interacts with a strong external error direction. The ensemble produces 1,039 standing-to-locomotion errors, representing 82.8% of its 1,255 image-level errors. Its locomotion recall is 0.915 for stand-co-tagged person rows and 0.977 for the 87 locomotion rows without a stand co-tag.

These counts do not show that co-tagging alone causes the standing errors. They show that the external label space mixes posture and action semantics that POLAR treats as separate exclusive states. The V-COCO result is best read as transfer under a partially incompatible annotation policy, not as a pure estimate of visual domain shift.

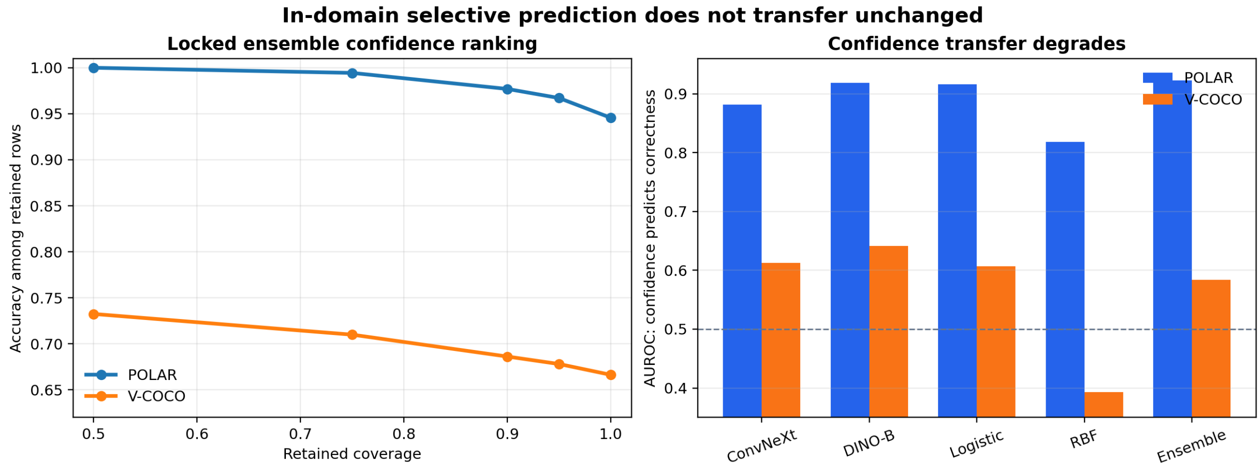
8.4 Post-hoc, non-selective multi-person analysis

There are 354 V-COCO images containing more than one mapped person label, covering 1,039 person rows. A full-frame ConvNeXt prediction is identical for every person in the same image by construction. It consequently shows zero within-image prediction variation and reaches 0.4793 person-level macro-F1 in these mixed scenes.

DINOv2-B top-four adaptation uses a person-conditioned crop. Its predictions differ between people in 171 of the 354 images and reach 0.6411 macro-F1. The post-hoc cluster-bootstrap difference from ConvNeXt is +0.1619 macro-F1 (95% interval [0.1300, 0.1943]) when images, rather than individual people, are resampled. DINOv2-S shows comparable behavior, varying in 175 mixed images and reaching 0.6416 macro-F1.

This is a functional difference between input pipelines, not evidence that transformer architecture alone is responsible. When an image contains people with different labels, a person-conditioned view can represent the evaluation unit, whereas a single full-frame prediction cannot.

8.5 Post-hoc, non-selective consensus and confidence under shift



Post-hoc confidence and shift diagnostics

On POLAR, components disagree on 15.7% of rows. The ensemble error rate is 2.0% when all components agree and 23.9% when they disagree. Disagreement has an error-detection AUROC of 0.782, and ensemble confidence has correctness AUROC 0.923.

On V-COCO, disagreement rises to 31.0%, but its diagnostic value falls. The ensemble error rate is already 28.5% on unanimous rows and 44.3% on disagreement rows; disagreement AUROC is 0.576 and confidence correctness AUROC is 0.584. Of 738 unanimous external errors, 684 (92.7%) are standing-to-locomotion.

The contrast is important. In-domain errors are often associated with recognizable model uncertainty or disagreement. Under external shift, many errors become shared and systematic, so neither confidence filtering nor a fixed ensemble can reliably identify them. At 90% confidence coverage, ensemble accuracy is 0.9770 on POLAR but only 0.6862 on V-COCO. The external figure is an exploratory diagnostic, not a validated rejection policy.

9. Attribution faithfulness

9.1 Protocol

The locked audit uses 256 deterministic test images, balanced across the four classes and four source-box-area quartiles. ConvNeXt-S is evaluated with Grad-CAM and DINOv2-B with 16-step integrated gradients. Both methods target the model's locked predicted class. Perturbation operates on an 8 by 8 grid with a nested random-deletion control.

The audit reports distinct questions rather than a single explanation score:

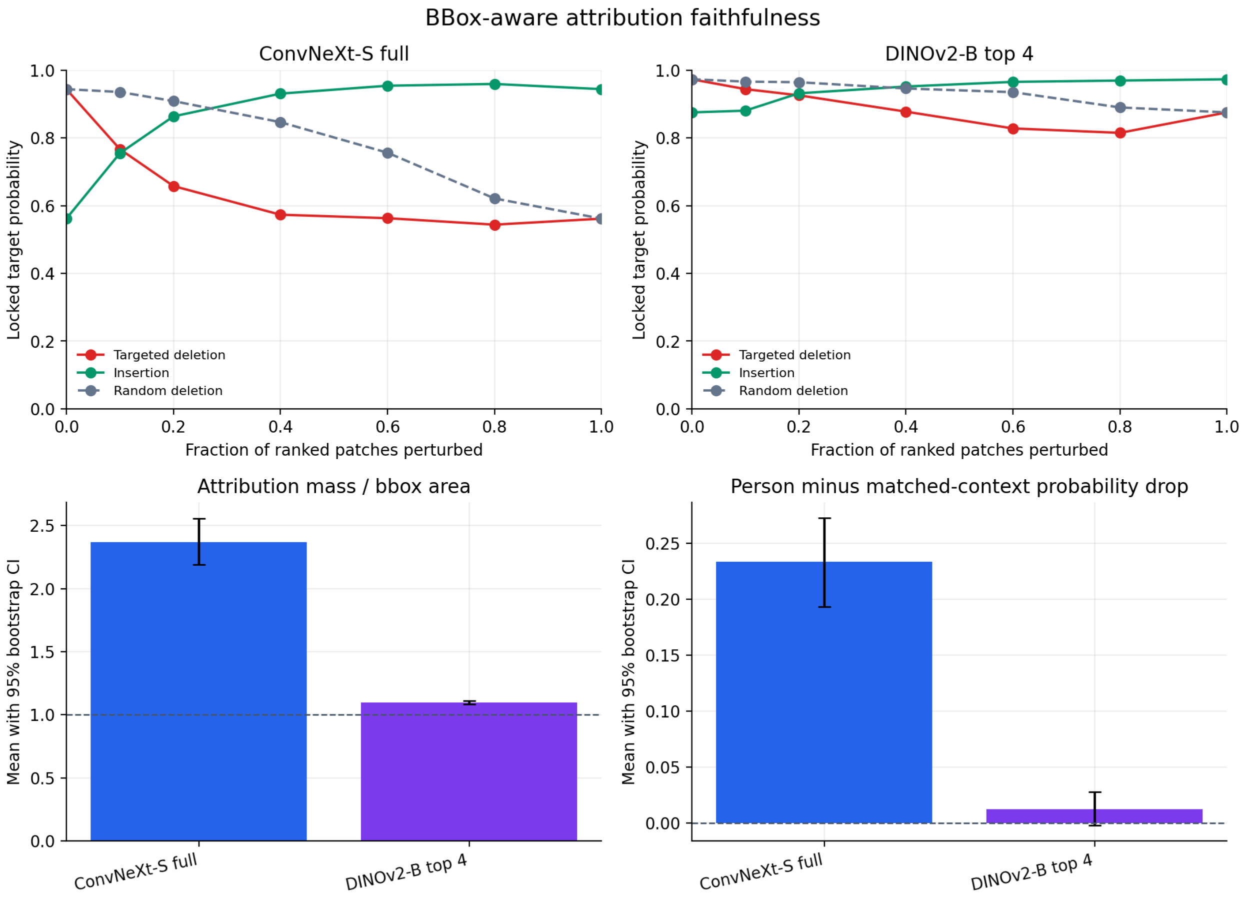
- deletion and insertion behavior;
- targeted deletion relative to a random-deletion baseline;
- raw attribution mass in the person box;
- attribution mass divided by box-area share;
- pointing-game localization;
- equal-area person versus context occlusion;

- consistency between full and cropped views;
- sensitivity to an alternative target class;
- sensitivity to randomized classifier and adapted-layer parameters.

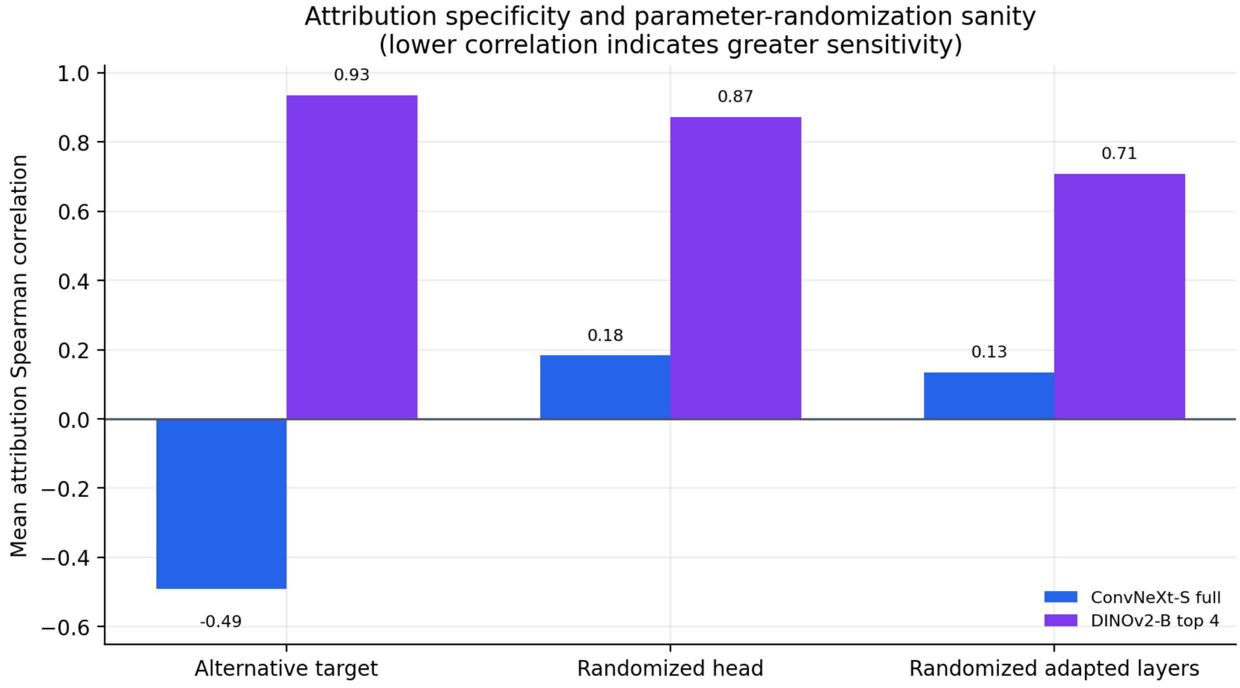
9.2 Predeclared auxiliary diagnostic results

Metric	ConvNeXt-S Grad-CAM	DINOv2-B integrated gradients
Deletion AUC, lower is better	0.615	0.874
Random deletion AUC	0.778	0.932
Random minus targeted deletion gap	**0.163**	0.058
Insertion AUC, higher is better	0.896	**0.946**
Person attribution mass	0.691	**0.848**
Person mass divided by box area	**2.368**	1.095
Pointing-game rate	**0.859**	0.828
Person minus context probability drop	**0.234**	0.012
Alternative-target Spearman rho	**-0.491**	0.934
Randomized-head Spearman rho	**0.183**	0.871
Randomized-adapted-layers Spearman rho	**0.135**	0.708

All entries are arithmetic means. The main attribution and perturbation metrics use 256 rows per family; matched person-context occlusion uses 254 because two rows per family have no equal-area context region; and each parameter-randomization correlation uses 16 rows per family.



Faithfulness perturbation and localization results



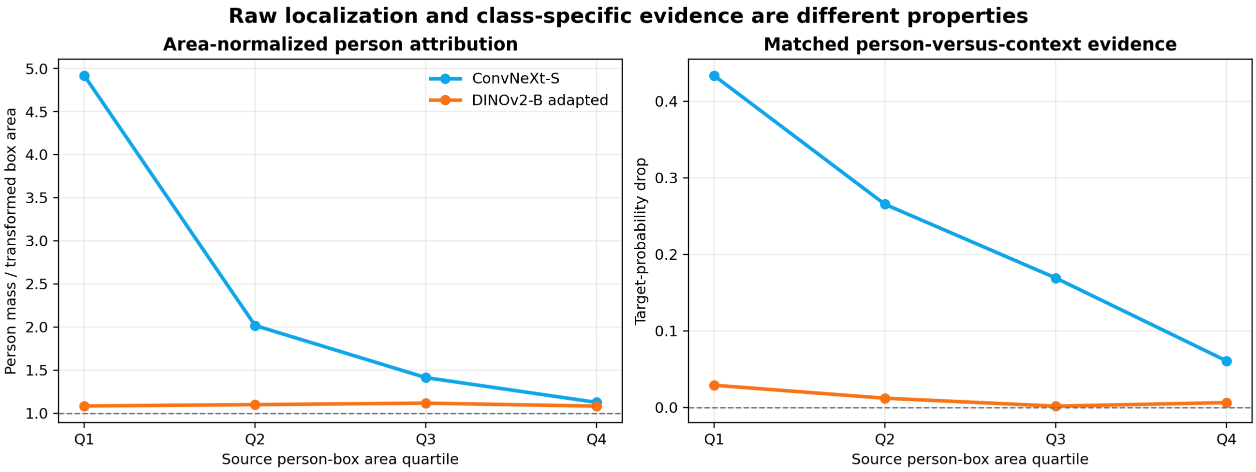
Attribution sanity checks

ConvNeXt provides the stronger evidence under this protocol. Its targeted deletion is more selective than random deletion, person attribution exceeds what box area alone would imply, person occlusion has a larger effect than matched context, and maps change substantially when the target or learned parameters change.

DINOv2-B integrated gradients place a high fraction of attribution inside the person crop and achieve high insertion AUC. Those results are insufficient by themselves: the person occupies much of the analyzed crop, equal-area person occlusion produces little probability change, and rank correlations remain high after target and parameter randomization. Under the tested baseline, resolution, and integration rule, the maps do not establish strong class-specific faithfulness. They may still indicate coarse person localization.

This is a comparison between complete explanation pipelines, not an architecture-controlled experiment. ConvNeXt uses a full-frame input and Grad-CAM; DINOv2 uses a person-context crop and integrated gradients. A different baseline, layer, attribution family, resolution, or perturbation operator could change the result.

9.3 Post-hoc, non-selective scale confounding



Post-hoc attribution-by-scale analysis

Raw person attribution mass increases with person-box size for both systems. For ConvNeXt, mean mass rises from 0.477 in the smallest quartile to 0.831 in the largest, while area-normalized lift falls from 4.916 to 1.125. The

Spearman correlations with box area are +0.543 for raw mass and -0.677 for lift. The matched person-minus-context probability drop also falls from 0.433 to 0.061.

For DINOv2, mean raw mass rises from 0.808 to 0.901, but lift remains near 1.08 to 1.12 and has no meaningful descriptive correlation with box area (rho = -0.043). Alternative-target correlations remain approximately 0.93 in every quartile, and matched person-context occlusion remains close to zero.

The analysis clarifies why raw localization mass should not be interpreted alone. A large box mechanically captures more attribution, and a person crop further constrains where pixels can fall. Normalized lift, matched-region perturbation, target sensitivity, and parameter randomization provide necessary counterchecks. The DINOv2 sanity-check result is not explained by one problematic size quartile.

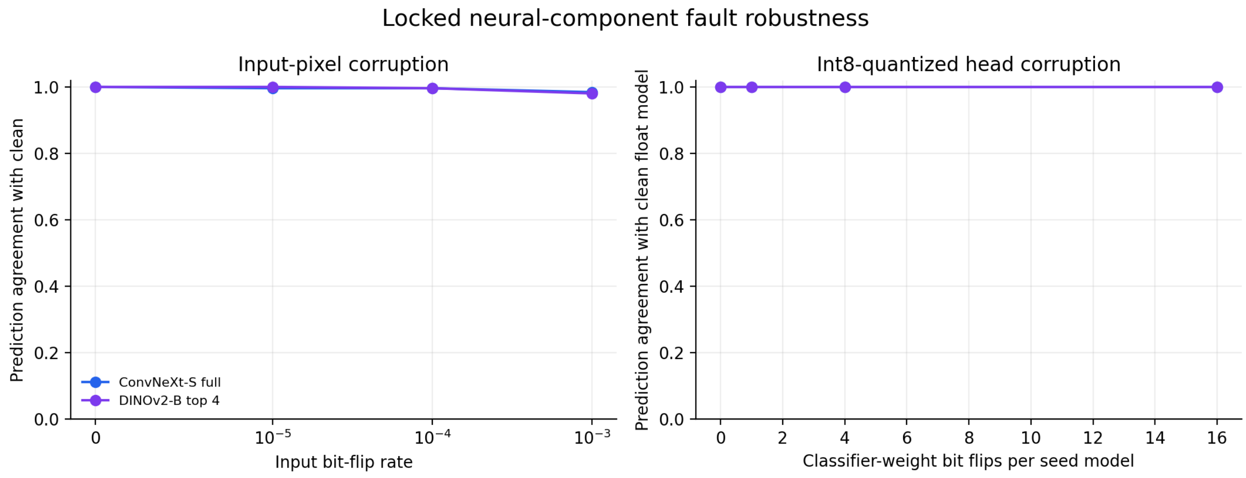
Two rows per model have projected person boxes covering the full analyzed frame, so no equal-area context region exists. They remain in every other metric and are excluded only from that matched-context statistic. The maximum recorded difference between recomputed attribution-path and locked probabilities is 0.00176.

10. Bounded fault response

10.1 Intervention

Input faults flip exact random bits in the post-resize uint8 RGB tensor before normalization. Parameter faults symmetrically quantize each classifier weight matrix to int8, flip exact bits, run inference, and restore the original float weights. Three declared fault seeds are averaged. The same 256-row cohort used by the faithfulness audit supports the diagnostic.

Condition	ConvNeXt-S agreement	DINOv2-B agreement
Input bit-flip rate 0.00001	0.996	1.000
Input bit-flip rate 0.0001	0.996	0.996
Input bit-flip rate 0.001	0.984	0.980
16 classifier-matrix flips per seed model	1.000	1.000



Bounded random bit-flip results

The models are locally stable under these specific random corruptions. At the highest input rate, ConvNeXt changes four of 256 labels and DINOv2 changes five.

10.2 Post-hoc, non-selective boundary analysis

All five DINOv2 label changes occur on true walking rows. Mean absolute probability drift is 0.0160 for walking, compared with 0.0026 for sitting, 0.0046 for standing, and 0.0017 for running. The median clean top-two margin is 0.0855 for changed DINOv2 predictions and 0.9999 for stable predictions. For ConvNeXt, the corresponding medians are 0.2178 and 0.9961.

Only five DINOv2 changes and four ConvNeXt changes are observed, so the class concentration is a hypothesis-generating pattern rather than a stable rate estimate. It nevertheless agrees with the locked confusion matrix: perturbations most often expose the same walking boundary that limits clean performance.

The experiment does not evaluate targeted progressive bit attacks, corruption in every backbone tensor, persistent faults, memory layout, hardware error rates, or recovery. Perfect agreement for the small classifier-matrix intervention is not proof of fault tolerance. It reports only what happened under the declared bounded intervention.

11. Integrated interpretation

This section is a post-hoc, non-selective synthesis of the locked and exploratory evidence. It does not revise the selected system or elevate exploratory findings to confirmatory status.

11.1 What produced the in-domain result

The evidence separates four sources of improvement.

First, the largest isolated change measured under a fixed pipeline is associated with additional labeled data. The frozen-representation curve gains 0.0664 macro-F1 from 242 to 9,958 rows. Walking and standing receive most of that improvement.

Second, representation and input view matter. DINOv2-B supports strong results through both selective adaptation and frozen multilayer features. ConvNeXt is weaker alone but provides distinct full-frame errors and stronger attribution sanity evidence.

Third, conservative regularization can prevent regressions or improve probability quality, but no single intervention explains the final score. Moderate augmentation gives the best DINOv2-S mean macro-F1; label smoothing gives the best calibration; larger dropout does not reliably dominate the baseline.

Fourth, probability averaging converts diversity into a locked gain. Seed averaging adds small increments, and combining five heterogeneous components adds 0.0125 macro-F1 over the strongest standalone RBF component. The equal-weight and leave-one-out checks suggest that broad component coverage matters more than exact weight precision.

11.2 What remains difficult

The remaining POLAR errors are overwhelmingly concentrated at adjacent posture and motion boundaries. This structure persists as training size grows. It also appears in the fault audit. The observation points toward a data and task limitation: a single frame does not always contain enough temporal evidence to separate standing from walking or walking from running.

The external results expose a second limitation. Person size changes, annotation semantics overlap, and full-frame models cannot distinguish multiple people in one image. More importantly, model errors become systematic: a large number of external standing examples are unanimously classified as locomotion. Ensembling independent seeds and model families cannot remove an error shared across all of them.

11.3 Defensible hypotheses for future work

The following hypotheses are supported enough to motivate new predeclared experiments, but not enough to be presented as established causal findings.

1 **Transition-aware labels or temporal input will help the walking boundary more than

additional generic regularization.** The same adjacent-state errors dominate across scale, the final ensemble, and faults.

2 **Person-conditioned views are necessary for person-level evaluation in mixed

scenes.** This follows from the evaluation unit: full-frame predictions cannot vary within an image, whereas crop-conditioned models can.

3 **External calibration requires shift-aware validation.** In-domain confidence and disagreement are informative, but both degrade sharply on V-COCO.

4 **Annotation policy is part of the domain.** Non-exclusive V-COCO action tags and exclusive POLAR state labels create a semantic shift that image augmentation alone cannot resolve.

5 **Attribution localization should be normalized for available area.** Raw person mass is strongly confounded by box size and crop construction; target and parameter randomization remain necessary.

6 **A compact linear probe is a credible deployment baseline.** The small in-domain RBF gain is unresolved by paired uncertainty and carries a large cost, although the nonlinear model can still add ensemble diversity.

Testing these hypotheses would require a new protocol, new selection budget, and a fresh held-out evaluation. None should be retrofitted into the present locked result.

12. Limitations and threats to validity

12.1 Dataset and split validity

The study covers four of nine POLAR classes. Its score must not be generalized to the full dataset. The source audit detects declared forms of related imagery but cannot recover unavailable subject or capture-session identities. Residual source relationships may remain below retrieval thresholds.

The external evaluation uses a forced mapping from non-exclusive V-COCO actions to exclusive posture states. This is useful for stress testing, but it is not a clean replication of the POLAR task. V-COCO is also not an independently collected posture benchmark designed for this label set.

12.2 Modeling validity

Architecture, view, and attribution method are partly coupled. ConvNeXt uses full frames; adapted DINOv2 uses person-context crops; frozen probes concatenate two views. Differences cannot be assigned to architecture alone.

The search is bounded rather than exhaustive. It does not cover all learning rates, crop ratios, augmentations, backbones, probe kernels, or blend structures. DINOv3 [26], ConvNeXt V2 [24], pose-specialized systems, ordinal losses, and temporal models are not tested and should not be discussed as though they were.

The scale curve contains five deterministic nested subsets. Repeated table rows at a given size do not provide independent seed variation. A genuine data-scaling study would draw multiple independent subsets at every size, preserve nesting where needed, and report variance.

12.3 Statistical validity

Locked uncertainty covers finite test-sample variation under class-stratified resampling. It does not cover uncertainty from dataset construction, undetected source overlap, alternative task mappings, or all possible training seeds.

Post-hoc analyses reuse the test and external predictions after observing aggregate results. Their intervals quantify sampling variation for fixed exploratory specifications; they do not restore confirmatory status. Multiple exploratory correlations are not corrected for multiplicity.

12.4 Explanations and faults

Attribution metrics are sensitive to the chosen baseline, layer, perturbation distribution, mask resolution, and crop. Passing the present checks would not prove causal human-like reasoning, and failing them does not prove that every use of a method is invalid.

Random input and classifier-matrix bit flips are bounded software interventions. They do not represent hardware reliability, adversarial robustness, end-to-end system safety, or faults elsewhere in the backbone.

12.5 Reproducibility boundary

The repository is designed to support traceability and retraining, not byte-for-byte replay of every local result. Raw images, trained checkpoints, fitted classifier binaries, dense per-example probabilities, and full-resolution attribution arrays remain local because of dataset rights, size, and privacy boundaries. Portable aggregate results and hashes allow consistency checks but cannot reconstruct excluded artifacts.

13. Reproducibility and availability

The repository targets Python 3.11. The reported run used PyTorch 2.11.0+cu126 and an NVIDIA GeForce RTX 4060 Laptop GPU. The tracked environment lock, protocols, source code, tests, lint configuration, and portable validation scripts define the public retraining surface. Continuous integration runs lint, compilation, tests, and the portable repository validator on Linux with CPU PyTorch.

The evidence package includes:

- locked train, validation, and test manifest hashes;
- the quarantine list and source-overlap audit;
- development training and failure ledgers;
- candidate selection, confirmation, classifier, and blend locks;
- per-seed and aggregate locked test metrics;
- class-stratified and paired bootstrap summaries;
- external image- and person-level metrics and overlap audit;
- faithfulness curves, randomization checks, and cohort details;
- fault definitions and aggregate outcomes;
- post-hoc exploratory tables and deterministic analysis code;
- a final evidence manifest containing hashes for portable locked artifacts.

Primary entry points are listed in Appendix B. Raw dataset access remains subject to the original POLAR and V-COCO terms. Versioned archives should attach the rendered report, preserve the citation metadata, and state the same artifact exclusions. When a DOI is assigned, it should resolve to an immutable release rather than to a moving development branch.

14. Conclusion

The locked five-component system achieves strong four-class POLAR performance: 0.9399 macro-F1 with a positive paired interval over every component. The result is supported by the source audit, development-only selection, hash-verified final fits, one-time test gate, and paired uncertainty, which constrain the usual routes by which a score can be inflated.

The broader study is more informative than the headline metric. The largest isolated gain observed under a fixed pipeline accompanies increased training-set size. Walking and standing remain the central boundary. A calibrated RBF SVM is marginally stronger than a logistic probe but far less efficient, and their paired difference is unresolved. Ensemble diversity matters more than precise weight tuning. Person scale contributes to external degradation, while annotation semantics and shared model bias explain why large errors remain even when the models agree. ConvNeXt Grad-CAM shows substantially greater target and parameter sensitivity under the tested sanity checks than DINOv2 integrated gradients. Random bit flips expose ambiguous boundaries but do not establish system reliability.

The report therefore supports a measured claim: this is a carefully controlled empirical benchmark and a reusable evaluation workflow for still-image posture recognition. It does not support a state-of-the-art, full-POLAR, architecture-only, deployment, causal-explanation, or hardware-robustness claim.

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Appendix A. Compact methods specification

A.1 Tasks and views

Item	Locked specification
Primary target	Four classes: sitting, standing, walking, running
Secondary target	Three classes: sitting, standing, walking/running
Development rows	9,958 train and 3,327 validation
Held-out rows	3,329 test
ConvNeXt view	Resized full frame
Adapted DINOv2 view	Person box plus 25% context
Frozen probe views	Full frame plus person box with 10% context
External units	3,761 unanimous-label images; 6,640 mapped people

A.2 Optimization controls

Control	Specification
Final neural seeds	42, 52, 62
Effective batch size	64 in the selected development recipes
Gradient clipping	1.0
Backbone learning rate	5e-6 in selected adaptation runs
Head learning rate	1e-3
Warmup fraction	0.10
Primary selector	Macro-F1
Practical tie	Absolute difference below 0.002
Bootstrap	10,000 class-stratified resamples
Locked bootstrap seed	20260822

Augmentation labels refer to repository-defined policies rather than universal standards. Exact transforms, normalization, layer decay, scheduling, and stopping behavior are encoded in the run requests and training source.

A.3 Attribution controls

Item	ConvNeXt-S	DINOv2-B
Method	Grad-CAM	Integrated gradients
Input	Full frame	Person plus 25% context
Target	Locked predicted class	Locked predicted class
Cohort	256 balanced rows	Same 256 rows
Perturbation grid	8 by 8	8 by 8
Random control	Nested random deletion	Nested random deletion
Sanity checks	Alternative target, randomized head, randomized adapted layers	Same

A.4 Fault controls

Item	Specification
Cohort	256 deterministic test rows
Input representation	Post-resize uint8 RGB before normalization
Input rates	0.00001, 0.0001, 0.001
Parameter representation	Symmetric int8 classifier matrices
Parameter intervention	16 flips per seed model
Repetitions	Three declared fault seeds
Restoration	Original float weights restored after inference

Appendix B. Evidence map

Question	Portable evidence
Data lock and counts	results/polar_data_lock.json
Quarantined source relations	results/polar_quarantine.csv
Training ledger and failures	results/polar_training_runs.csv; results/polar_training_failures.json
Multi-seed confirmation	results/polar_confirmation_summary.json
Classifier selection	results/polar_classifier_summary.json
Ensemble weights	results/polar_validation_blend.json
Final selection lock	results/polar_final_selection_lock.json
Test access gate	results/polar_test_access_gate.json
Locked metrics	results/polar_test_metrics.csv
Per-class test results	results/polar_test_per_class.csv
Locked uncertainty	results/polar_test_uncertainty.json
External evaluation	results/polar_external_summary.json
Attribution audit	results/polar_faithfulness_summary.json
Fault audit	results/polar_fault_summary.json
Post-hoc analyses	results/polar_exploratory_summary.json and polar_exploratory_*.csv
Portable artifact hashes	results/polar_final_evidence_manifest.json

Appendix C. Confirmatory and exploratory boundary

Result	Role
Locked four-class ensemble score	Confirmatory held-out evaluation
Paired component comparisons in the locked uncertainty file	Confirmatory
Collapsed three-class test score	Locked secondary evaluation
V-COCO aggregate transfer table	Locked external audit
Faithfulness method and cohort	Locked before test attribution execution
Declared bit-flip conditions	Locked bounded diagnostic
Adaptation-efficiency and regularization syntheses	Post hoc, non-selective development evidence
Seed-average versus mean-seed comparison	Post hoc, non-selective
Error adjacency decomposition	Post hoc, non-selective
Box-area quartiles and geometry baseline	Post hoc, non-selective
Equal-weight and leave-one-out blends	Post hoc sensitivity analysis
Adapted-versus-frozen paired pipeline comparisons	Post hoc, non-selective
Confidence coverage and disagreement under shift	Post hoc, non-selective
V-COCO co-tag semantics	Post hoc, non-selective
Mixed-person within-image behavior	Post hoc, non-selective
Attribution scale correlations	Post hoc, non-selective
Fault changes by class and margin	Post hoc, non-selective

The exploratory analyses are retained because they expose mechanisms and limitations. Their role labels prevent them from being mistaken for a second round of model selection on the held-out data.